

Assignment 3.4

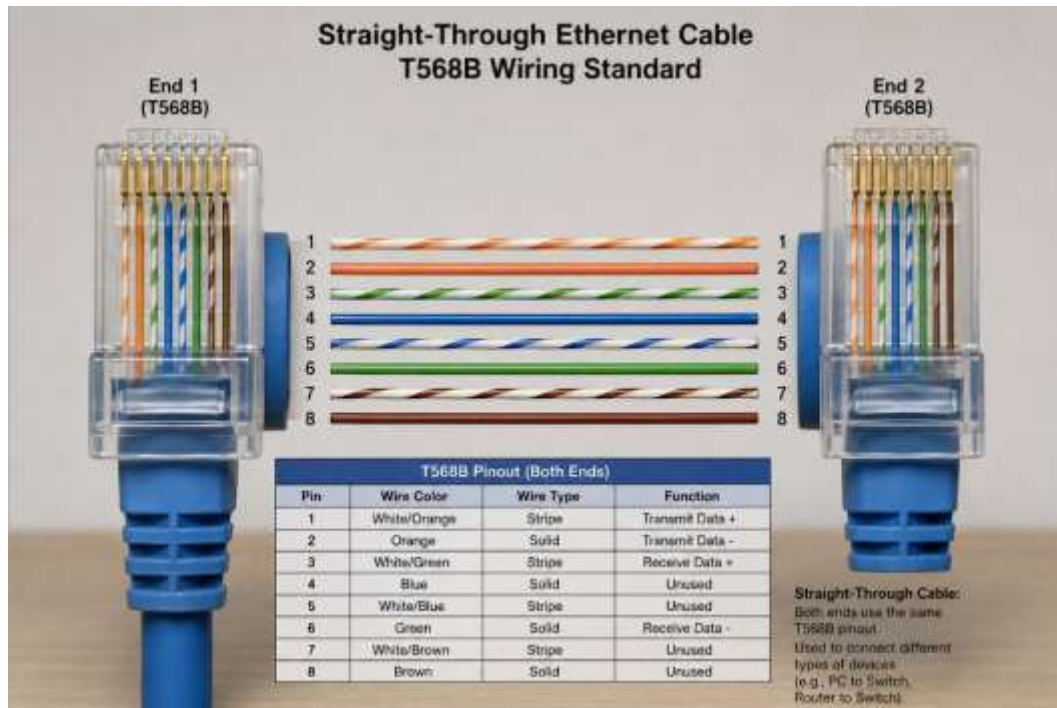
1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Difference	Ethernet Cable	Fiber Optic Cable
1. Transmission medium	Uses copper wires to transmit electrical signals.	Uses glass/plastic fibers to transmit light signals.
2. Speed & distance	Suitable for shorter distances, commonly up to 100 m for standard copper Ethernet.	Supports much longer distances and very high data speeds.
3. Interference	Can be affected by electromagnetic interference (EMI).	Resistant to EMI because it uses light instead of electrical signals.

Real-world examples

- **Ethernet cable:** Connecting a home Wi-Fi router to a desktop PC, gaming PC, or switch.
- **Fiber optic cable:** Connecting different buildings in a college campus backbone or connecting servers inside a data center.

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



- White/Orange
- Orange
- White/Green
- Blue
- White/Blue
- Green
- White/Brown
- Brown

3. **Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.**

Step-by-step guide

1. **Cut the cable:**

Cut the CAT6 cable to the required length using the crimping tool.

2. **Remove the outer cover:**

Use the cable stripper to carefully remove about 2–3 cm of the outer jacket from the end of the cable.

3. **Separate the wires:**

Separate the 8 small colored wires inside the cable.

4. **Arrange the wires:**

For the T568B standard, arrange the wires in this order:

White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.

5. **Straighten the wires:**

Straighten all 8 wires and keep them next to each other in the correct order.

6. **Cut the wires evenly:**

Trim the wires so that all 8 ends are the same length.

7. **Insert into RJ45 connector:**

Carefully push the wires into the RJ45 connector. Make sure each wire reaches the correct pin and goes all the way inside.

8. **Check the order:**

Look at the connector and confirm that the wires are still in the correct T568B order.

9. **Crimp the connector:**

Put the RJ45 connector into the crimping tool and press it firmly. This locks the wires into the connector.

10. **Test the cable:**

Use an Ethernet cable tester to check whether the cable is working correctly.

T568B Color Order

1 → White-Orange

2 → Orange

3 → White-Green

4 → Blue

5 → White-Blue

6 → Green

7 → White-Brown

8 → Brown

- 4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.**

Hint: Consider speed, distance, and budget for your answer.



I would use Ethernet (CAT6) cables to connect the 10 gaming PCs because they provide good speed and are suitable for short distances. Ethernet cables are also cheaper and easier to install than fiber optic cables. I would use an RJ45 crimper to attach RJ45 connectors to the CAT6 cables. Fiber optic cables and a fiber cleaver are not necessary for a small gaming café because they would cost more and are mainly useful for long-distance or high-speed backbone connections.